

Unit:1 Introduction to Big Data and Hadoop

1. Types of Digital Data
2. Introduction to Big Data
3. Big Data Analytics
4. Role of Distributed System in Big Data
5. Current Trends in Big Data Analytics
6. Structure and Unstructured Data
7. Introduction to Nosql databases

Digital Data and its Types

1. **Digital Data** refers to information that is stored, processed, and transmitted using binary code (0s and 1s). It is the basis of all modern computing systems, allowing for the storage and manipulation of data by electronic devices.

Types of Digital Data

Digital data can be categorized into the following main types:

1. Text Data

Represents letters, numbers, and symbols.

Encoded using formats like ASCII, Unicode.

Examples: Emails, documents, source code.

Digital Data and its Types

2. Numeric Data

Includes integer and floating-point numbers.

Used in mathematical operations and calculations.

Examples: Age, salary, temperature readings.

3. Audio Data

Represents sound in digital form.

Encoded using formats like MP3, WAV, AAC.

Used in music files, voice recordings.

4. Image Data

Visual data stored as a grid of pixels.

Can be raster (JPEG, PNG) or vector (SVG).

Used in photography, graphic design, web content.

Digital Data and its Types

5. Video Data

A sequence of images (frames) with sound, stored digitally.

Formats include MP4, AVI, MKV.

Used in films, animations.

6. Graphical Data

Includes diagrams, drawings, and user interface elements.

Often used in CAD software, games, and design applications.

7. Binary Data

General category for any non-text files.

Includes executable files, software programs, compressed files (e.g., ZIP, EXE).

Introduction to Big Data

What is Big Data?

Big Data refers to extremely large and complex sets of data that traditional data processing tools and techniques cannot handle efficiently. It includes the structured, semi-structured, and unstructured data generated at high speed from various sources like social media, sensors, transactions, and devices.

- Big data is a term applied to a new generation of software, applications, and system and storage architecture.
- It designed to provide business value from unstructured data.
- Big data sets require advanced tools, software, and systems to capture, store, manage, and analyze the data sets.

Who uses big data?

Facebook, Amazon, Google, Yahoo, New York Times, twitter and many more

Introduction to Big Data

Key Characteristics of Big Data (The 5 V's)

Volume – Refers to the massive amount of data generated.

Example: Facebook generates terabytes of data daily.

Velocity – The speed at which data is generated and processed.

Example: Streaming data from sensors or social media feeds.

Variety – The different types of data formats.

Structured (tables), semi-structured (XML, JSON), unstructured (videos, emails).

Veracity – The accuracy and trustworthiness of the data.

Data can be inconsistent or incomplete.

Value – The potential insights and benefits that can be drawn from the data.

Extracting actionable insights from big data can drive business decisions.

Introduction to Big Data

Sources of Big Data

Social Media (e.g., Twitter, Facebook)

IoT Devices and Sensors

Business Transactions

Web and Mobile Applications

Health and Medical Records

Importance of Big Data

Helps in better decision-making

Enables real-time analytics and predictions

Improves customer experience and personalization

Supports innovation in fields like healthcare, finance, and logistics

Introduction to Big Data

Big Data Technologies

Storage: Hadoop Distributed File System (HDFS), Amazon S3

Processing: Apache Hadoop, Apache Spark

Analytics: Power BI, Apache Hive

Databases: MongoDB, HBase

Big Data Analytics

- **Big Data Analytics** means **studying or analyzing a very large amount of data** to find useful information.
- It is the process of **examining large and complex data sets** (called *big data*) to discover **useful patterns, trends, relationships, and insights** that help in making better business, scientific, or social decisions.
- These data sets are **too large** or **too complex** to be handled by traditional data-processing tools like Excel or regular databases.

Why is Big Data Analytics Important?

- Big data analytics helps:
- Improve **business decisions**
- Understand **customer needs**
- Detect **fraud** and **security threats**
- Optimize **operations** and **supply chains**
- Predict **future outcomes** (using predictive analytics)
- Personalize **marketing and customer experience**

Big Data Analytics

- Technologies Used in Big Data Analytics

Data Storage Tools

Hadoop Distributed File System (HDFS)

NoSQL Databases like MongoDB

Data Processing Tools

Apache Hadoop: Batch processing of huge data sets

Apache Spark: Faster, in-memory data processing

Data Analysis Tools

Python and **R:** Popular programming languages for analytics

SQL: For querying structured data

Visualization Tools

Power BI

Google Data Studio

Big Data Analytics

- Steps in Big Data Analytics Process

Data Collection

Gather data from various sources: websites, sensors, social media, devices, logs, etc.

Data Storage

Store the data using scalable platforms like cloud services, Hadoop, or NoSQL databases.

Data Cleaning & Preparation

Remove errors, duplicates, and irrelevant data; transform it into useful formats.

Data Analysis

Use tools and models to find patterns, trends, and insights.

Data Visualization

Create charts, graphs, dashboards to present findings clearly.

Decision Making

Use the results to make data-driven decisions.

Big Data Analytics

- **Real-Life Examples of Big Data Analytics**
- **E-commerce:** Amazon uses big data to recommend products based on your browsing and buying history.
- **Healthcare:** Hospitals analyze patient data to predict diseases early.
- **Finance:** Banks detect unusual transactions using real-time analytics.
- **Smart Cities:** Traffic patterns are analyzed to reduce traffic jam.
- **Education:** Student performance data is analyzed to improve teaching methods.

Role of Distributed System in Big Data Analytics

- **What is a Distributed System?**

A **distributed system** is a network of multiple computers (called nodes) that work **together as a single system** to perform tasks. These computers **share data, processing power, and storage**, often located in different physical locations.

- **Why are Distributed Systems Important in Big Data?**

Big Data involves **huge volumes of data** that are too large to be handled by a single machine. Distributed systems solve this by:

- **Dividing the work** across many machines
- **Storing data** across different servers
- **Processing data in parallel** for faster results

Role of Distributed System in Big Data Analytics

- Roles of Distributed Systems in Big Data

Role	Description
1. Data Storage	Stores data across multiple machines using tools like HDFS (Hadoop File System)
2. Parallel Processing	Processes big data faster by dividing tasks across nodes (e.g., MapReduce)
3. Scalability	Easily adds more machines to handle growing data volumes
4. Fault Tolerance	If one machine fails, others take over (no data loss or downtime)
5. High Availability	Data and services are always accessible, even if some servers go down
6. Load Balancing	Distributes workload evenly across systems to avoid overloading a single node
7. Real-Time Data Processing	Supports fast, live analytics using tools like Apache Spark

Current Trends in Big Data Analytics

1. **AI-Powered Analytics**

Artificial Intelligence and Machine Learning are being deeply integrated into analytics platforms to automate pattern detection, predictions, and decision-making.

2. **Cloud-Based Analytics**

Organizations are rapidly shifting to cloud platforms like AWS, Azure, and Google Cloud for scalability, cost-efficiency, and faster deployment. It allows storage and processing of massive data without maintaining physical infrastructure.

3. **Real-Time Analytics**

Real-time data processing tools are enabling instant insights and actions, especially in areas like fraud detection, recommendation systems, and live dashboards

4. **Data Privacy & Governance**

With increasing regulations, businesses are focusing more on managing data access and security through proper governance frameworks and tools.

Structure and Unstructured Data

Structure Data :

Data that is organized into predefined models like rows and columns (e.g., tables in databases).

Examples:

- Relational databases (SQL)

- Excel spreadsheets

- Employee records (Name, Age, Salary)

Characteristics:

- Highly organized

- Easy to search and analyze

- Stored in relational databases

- Uses schema (data types and formats predefined)

Tools Used:

- SQL, MySQL, Oracle, Microsoft SQL Server

Structure and Unstructured Data

Unstructure Data :

Data that has no fixed format or structure and is not organized in a predefined way.

Examples:

Emails

Videos, images

Social media posts

PDF documents

Chat logs

Characteristics:

No specific format

Harder to search and process

Requires advanced tools (e.g., AI, NLP)

Grows faster than structured data

Tools Used:

Hadoop, NoSQL (MongoDB, Cassandra)

Text mining tools, NLP engines

Introduction to Nosql databases.

- **NoSQL (Not Only SQL)** databases are a modern class of database management systems designed to handle **large volumes of diverse, rapidly changing data** that traditional relational databases (RDBMS) struggle with.

- Unlike SQL databases, which store data in structured tables with fixed schemas, NoSQL databases offer **flexible schemas** and are optimized for **scalability, speed, and performance**—especially in **big data** and **real-time web applications**.

- **Types of NoSQL Databases**

- **Document-based** – stores data as JSON-like documents
- **Key-Value** – simple data as key-value pairs
- **Column-based** – data stored in columns for fast access
- **Graph-based** – best for connected data like social networks